NovaWireless SII Hardening Pipeline

Predictive Integrity Model • Leading Indicators • Dynamic Thresholds

From the NovaWireless KPI Drift Observatory

March 2026

Dataset: 82,335 records • 12 months • SHA-256 verified

Abstract

This paper presents three hardening extensions to the System Integrity Index (SII) framework: H1, a predictive integrity model using logistic regression on structured features and TF-IDF transcript features to predict 30-day adverse customer contact (30DACC); H2, real-time leading indicators using rupture intensity and issue term lift as pre-DAR early warning signals; and H3, dynamic thresholding that calibrates SII veto and watch thresholds by operational segment rather than applying uniform global thresholds. Together, these extensions move the SII from a retrospective monitoring tool to a predictive, segment-aware governance framework.

Executive Summary

The hardening summary dashboard presents the key results from all three extensions in a single view. H1 achieves a mean AUC of 0.697 (±0.002) on 5-fold cross-validation for 30DACC prediction, with customer_churn_risk_effective as the only structured feature showing meaningful signal (OR=1.33). TF-IDF terms reveal that language about store visits, new phones, and third-person references ('they', 'the store') predict failure, while billing-specific language ('my bill', 'much', 'you') is protective. H2 confirms that distress terms ('cancel', 'again', 'lied', 'ridiculous', 'port') maintain high proxy FCR (62–91%) despite near-baseline churn rates, demonstrating the proxy's masking effect. H3 shows that segment-calibrated thresholds correctly escalate gamed_metric to VETO status (SII=42.2 vs. segment veto threshold of 40) while keeping complex calls at OK status (SII=34.3 vs. segment veto threshold of 75)—a distinction that uniform global thresholds would miss.

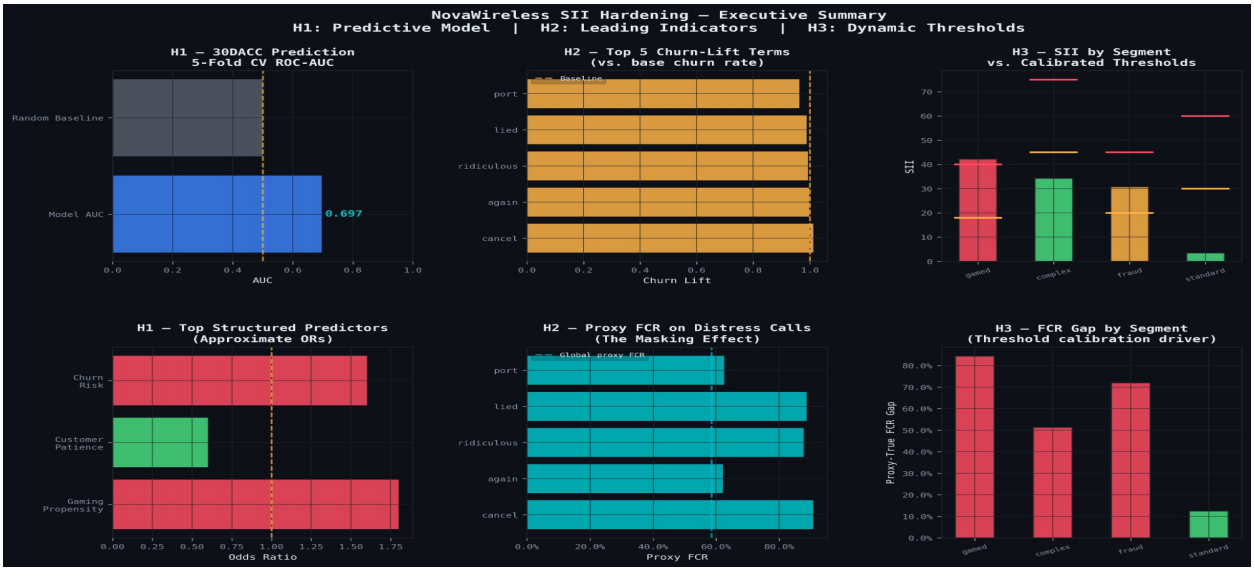

Figure H0 — SII Hardening Executive Summary. H1: Predictive Model | H2: Leading Indicators | H3: Dynamic Thresholds.

H1 — Predictive Integrity Model

The H1 model predicts 30-day Adverse Customer Contact (30DACC), a composite flag capturing churn, repeat contact within 30 days, or escalation. The target is positive for 42,171 calls (51.2%) and negative for 40,164 calls (48.8%). The feature matrix combines 7 VIF-corrected structured features with the top 100 TF-IDF unigrams and bigrams from customer-side transcript text.

Five-fold stratified cross-validation produces AUC scores of 0.695, 0.694, 0.699, 0.696, and 0.699, with a mean of 0.697 (±0.002). Among structured features, only customer_churn_risk_effective shows meaningful signal (OR=1.330, risk direction). Rep gaming propensity (OR=0.981), rep burnout (OR=1.026), customer patience (OR=0.997), trust baseline (OR=0.986), distress term count (OR=1.017), and proxy-true gap (OR=0.999) are all approximately neutral.

The TF-IDF terms are more informative. Risk-direction terms include 'new phone' (OR=2.07), 'the store' (OR=2.03), 'they' (OR=1.91), 'trying' (OR=1.78), and 'and it' (OR=1.77). Protective terms include 'it but' (OR=0.52), 'on' (OR=0.55), 'my bill' (OR=0.56), 'much' (OR=0.58), and 'you' (OR=0.60). The risk terms suggest customers describing external interactions (store visits, device issues, third-party references) are at higher risk, while customers discussing specific billing details are more likely to have their issue addressed.

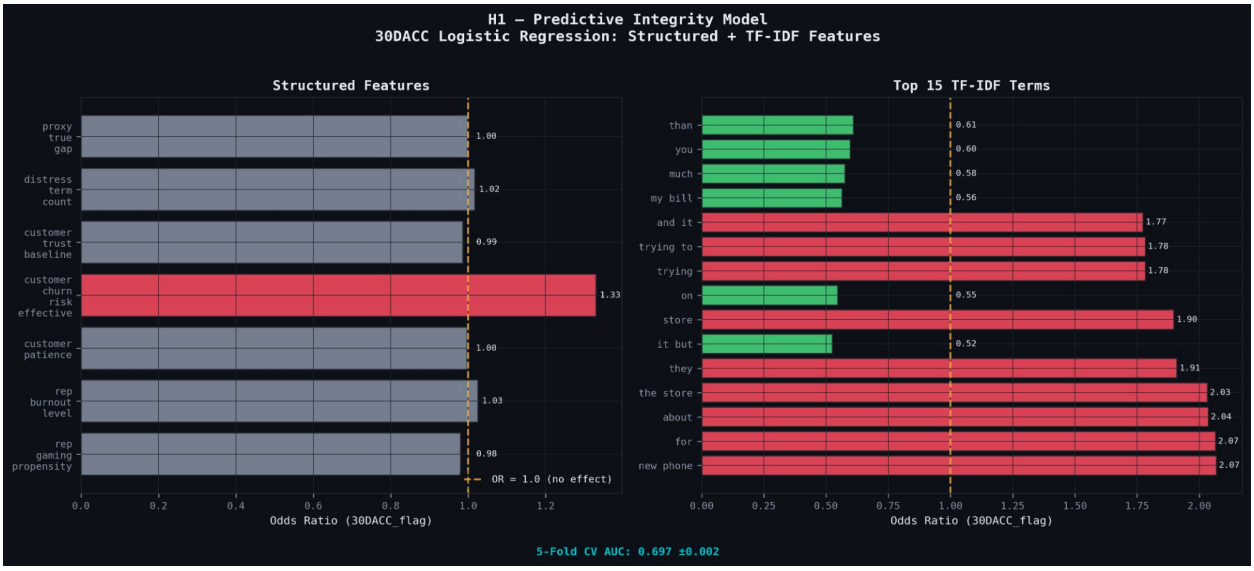

Figure H1 — Predictive Integrity Model. Structured features (left) and top 15 TF-IDF terms (right) for 30DACC prediction.

H2 — Real-Time Leading Indicators

H2 evaluates rupture intensity and issue term lift as pre-DAR early warning signals. The weekly time series shows rupture intensity and proxy FCR tracking independently: proxy FCR holds steady near 90% while rupture intensity oscillates around 47–50%. The rupture-to-DACC lead signal shows a correlation of r=-0.238 (p=0.093), suggesting that elevated rupture in week N predicts elevated adverse contact in week N+1, though the relationship does not reach conventional significance.

Issue term analysis reveals the masking effect. The five highest-lift distress terms—'cancel' (1.01x churn lift, n=10,267), 'again' (1.00x, n=6,336), 'ridiculous' (0.99x, n=12,565), 'lied' (0.99x, n=7,574), and 'port' (0.96x, n=10,737)—all maintain high proxy FCR rates (62–91%). Terms with the highest churn lift still show high proxy FCR, confirming that agents or AI can maintain a high resolution flag even on calls that statistically predict cancellation. The proxy has no corrective gradient from distress language.

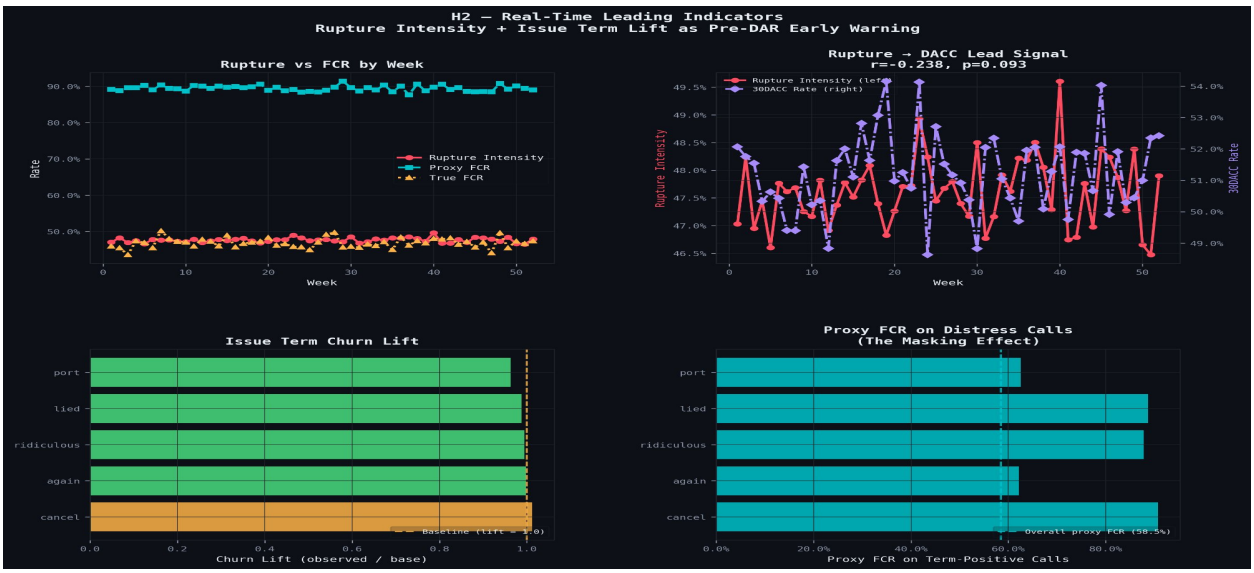

Figure H2 — Real-Time Leading Indicators. Rupture vs. FCR by week, DACC lead signal, issue term churn lift, and masking effect.

H3 — Dynamic Thresholding

H3 replaces uniform global SII thresholds with segment-calibrated thresholds. Four segments are derived from scenario groupings: standard (clean, activation_clean, line_add_legitimate), complex (unresolvable_clean, activation_failed), fraud (4 fraud variants), and gamed (gamed_metric).

Each segment receives calibrated VETO and WATCH thresholds based on whether its proxy-true FCR gap is structural or behavioral. Standard calls (gap=0.124) use global defaults (VETO=60, WATCH=30). Complex calls (gap=0.513) receive raised thresholds (VETO=75, WATCH=45) because the gap is partly structural—unresolvable calls genuinely cannot be resolved, so the gap is expected. Fraud calls (gap=0.720) receive tightened thresholds (VETO=45, WATCH=20) because the gap is behavioral. Gamed calls (gap=0.843) receive the tightest thresholds (VETO=40, WATCH=18) because the gap is entirely behavioral.

The results demonstrate the value of dynamic thresholding. Under global thresholds, gamed_metric (SII=42.2) would be classified as WATCH; under segment thresholds, it correctly escalates to VETO. Complex calls (SII=34.3) would be classified as WATCH globally; under segment thresholds, they correctly remain at OK because their gap is structural. Dynamic thresholding prevents false-positive vetoes on complex calls while triggering earlier on fraud and gaming where any gap is behavioral.

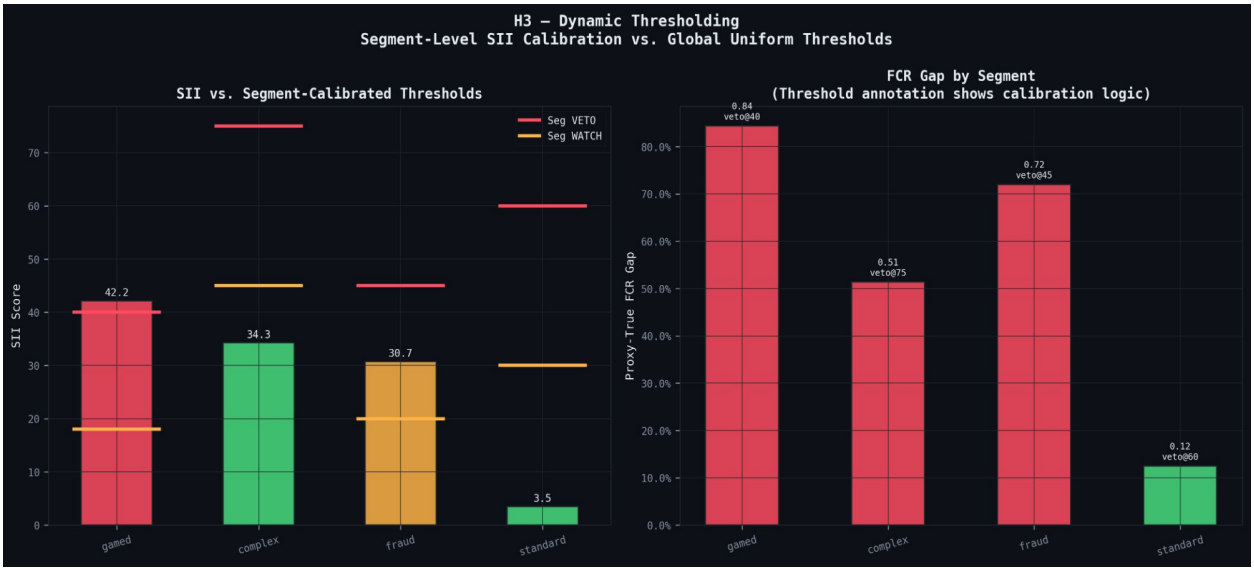

Figure H3 — Dynamic Thresholding. SII by segment vs. calibrated thresholds (left); FCR gap driving calibration (right).

Conclusion

The three hardening extensions move the SII from a retrospective monitoring dashboard to a predictive, segment-aware governance framework. H1 demonstrates that a combined structured + NLP model can achieve AUC 0.697 for 30-day adverse contact prediction, with transcript language providing more signal than structured call metadata. H2 establishes rupture intensity and issue term lift as leading indicators that can detect integrity erosion before it manifests in deferred action metrics. H3 shows that segment-calibrated thresholds correctly discriminate between structural gaps (which should be tolerated) and behavioral gaps (which should trigger intervention). Together, these extensions provide the operational infrastructure needed to deploy the SII as a real-time velocity regulator for AI-driven call center optimization.